REVISION 2026 · KERALA POLYTECHNIC

📘 Theory · 🕒 60 hrs · 🎓 4 Credits · 🏷️ 2012

Basic Surveying

Chain & Tape · Compass · Levelling · Theodolite — fundamental principles for civil engineering measurement and mapping.

Programme	Civil Engineering · Civil & Rural · Civil & Environmental · Construction Technology
Semester	II
Type	Theory
L:T:P:S	4:0:0:0
CIE | ESE	40 | 60

📄 Official Source Declaration

Source: Kerala Polytechnic Revision 2026 syllabus — Course 2012 Basic Surveying. All module topics, subtopics, learning outcomes, and assessment patterns are extracted from the official PDF. The handbook preserves the original structure and terminology.

Verified Inconsistencies: The PDF lists 4 modules with 60 instructional hours (Module I, II, III: 16 hrs each; Module IV: 12 hrs). Self-learning activities are listed as 6 hours per week (total 90 hrs) but the table in the PDF has a blank total row — we present the activities as given. The CO-PO mapping table has a total row that sums to 11, 9, 8, 6, etc. — these are presented as in the source.

📊 Course Dashboard

60
Contact Hours

4
Credits

40+60
CIE + ESE

4
Modules

L	T	P	S	SS/Week	Total Hrs	Credits	CIE	ESE
4	0	0	0	6	60	4	40	60

Teaching & Assessment Scheme

Pre-requisite: Basic mathematics (1002) & Fundamentals of building construction (1011).

🧭 How to Use This Handbook

1. Understand
Read each module’s introduction and topic details. Use the Malayalam notes for clarity.

2. Visualise
Study the diagrams, formula cards, and worked examples. Use the visual library.

3. Apply
Solve the numerical problems, use the calculators, and attempt the Q&A section.

4. Revise
Use the quick revision cards, glossary, and model paper for exam preparation.

🎯 Course Outcomes & CO-PO Mapping

CO	Description	Bloom Level
CO1	Develop competency in chain and tape surveying techniques and simple area calculations.	Understand
CO2	Compute reduced levels using standard levelling methods.	Apply
CO3	Solve numerical problems related to levelling applications and contour basics.	Apply
CO4	Explain basic theodolite concepts and methods of horizontal angle measurement.	Understand

Course Outcomes (CO) with Bloom's Taxonomy Level

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	2	1	-	-	-	-	-	-	-
CO2	3	3	2	2	1	-	-	-	-	-	-
CO3	3	3	2	1	1	-	-	-	-	-	-
CO4	2	1	2	2	-	-	-	-	-	-	-

CO-PO Articulation Matrix (Legends: 1-Low, 2-Medium, 3-High)

🗺️ Curriculum Coverage Map

Module	Topics	Hours	CO	Anchor
I	Chain & Tape, Compass, Plane Table (Intro)	16	CO1	Module 1
II	Levelling – Principles, Instruments, Reduction	16	CO2	Module 2
III	Levelling Applications, Contour Basics	16	CO3	Module 3
IV	Theodolite Surveying – Intro & Angle Measurement	12	CO4	Module 4

Module-wise topics, hours, CO mapping and handbook anchor

🔗 Module Roadmap

📏 **Module I**
Chain & Tape · Compass · Plane Table — fundamentals of distance and direction measurement.

📐 **Module II**
Levelling Principles — instruments, staff, temporary adjustments, reduction of levels (HI & Rise-Fall).

📈 **Module III**
Levelling Applications — fly, profile, reciprocal; contouring basics.

🔭 **Module IV**
Theodolite — parts, adjustments, horizontal angle measurement (repetition & reiteration).

MODULE I

Chain & Tape Surveying, Compass and Plane Table

16 hrs · CO1 · Bloom: Remember, Understand, Apply

Introduction to basic surveying instruments, station selection, ranging, offsetting, and area calculation. Compass survey – bearings, included angles, local attraction. Plane table survey – introduction, advantages and limitations.

▸Chain & Tape Instruments and Accessories

Chain: A chain is a measuring device made of galvanised mild steel wire, consisting of links joined by rings. Common types: 20m chain (100 links), 30m chain (150 links), and 100ft chain. **Tapes:** Cloth tape, steel tape, linen tape, and invar tape (used for high precision).

Fig: Chain with links and rings.

> **Malayalam support:** ചെയിൻ (Chain) അളക്കുന്നതിനുള്ള ഒരു ഉപകരണമാണ്. ഇത് ലോഹ വളയങ്ങളാൽ ബന്ധിപ്പിച്ച ലിങ്കുകൾ ഉൾക്കൊള്ളുന്നു. ടേപ്പുകൾ (Tapes) വിവിധ തരത്തിലുണ്ട്: സ്റ്റീൽ, ക്ലോത്ത്, ഇൻവാർ എന്നിവ.

▸Survey Lines – Base line, Tie line, Check line

Base line: The longest and most important line in a survey, usually run through the centre of the area. All other measurements are referred to it. **Tie line:** A line joining two points on the main survey lines to locate interior details. **Check line:** A line measured to verify the accuracy of the survey (also called a proof line).

> 💡 **Tip:** The base line should be laid on level ground and be easily measurable. Check lines are essential for error detection.

▸Ranging – Direct and Indirect

Ranging: The process of establishing intermediate points on a straight line between two survey stations. **Direct ranging:** When the ends are intervisible, intermediate points are fixed by line of sight. **Indirect ranging:** When ends are not intervisible (due to obstruction), the line is set out using random line or reciprocal ranging.

> **Worked example:** Two stations A and B are 200 m apart but a hill blocks the view. Using indirect ranging, a point C is chosen such that AC and BC are clear. Measure AC=80m, BC=120m. By reciprocal ranging, the line is transferred. (Detailed procedure in textbooks.)

▸Chaining and Taking Offsets

Chaining: Measuring distance along the chain line using a chain or tape. **Offsets:** Lateral measurements taken from the chain line to locate objects (e.g., buildings, boundaries). Types: *Perpendicular offset* (using cross-staff or optical square) and *oblique offset* (measured at an angle).

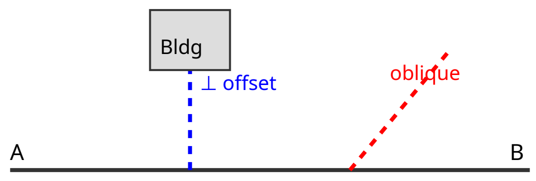


Fig: Perpendicular and oblique offsets from chain line.

▸Conventional Signs in Surveying

Standard symbols used on survey maps to represent features like buildings, roads, trees, water bodies, bridges, etc. These are defined by the Survey of India and other standards.

Feature	Symbol
Triangulation station	▲
Road	=====
River/Stream	~~~~~
Building	□
Tree	●

Common conventional signs

▸Simple Area Calculation

Areas can be computed using: (i) Mid-ordinate rule, (ii) Average ordinate rule, (iii) Trapezoidal rule, (iv) Simpson's rule. For simple figures, use geometry (rectangle, triangle, trapezium).

> **Worked example:** A field is measured with offsets at intervals of 10m: 2.5, 3.8, 4.2, 3.0, 1.8 m. Using Trapezoidal rule: Area = $d \times [(O_0+O_n)/2 + \Sigma O_i] = 10 \times [(2.5+1.8)/2 + (3.8+4.2+3.0)] = 10 \times [2.15 + 11.0] = 131.5$ m^2.

▸Compass Survey – Bearings, Meridians, Included Angles

Meridian: Reference direction. *True meridian* (geographic north) and *Magnetic meridian* (direction of magnetic needle). **Bearing:** Horizontal angle between a line and a meridian. *Fore bearing* (FB) and *Back bearing* (BB) differ by 180°. *Whole circle bearing* (WCB) measured clockwise from north (0°–360°). *Quadrantal bearing* (QB) measured from north or south towards east or west (0°–90°).
Included angle: The angle between two lines at a station. If bearings are known, included angle = BB of previous line − FB of next line (or vice versa, adjusted).

> **Worked example:** The FB of line AB = 60°30′, BB of line AB = 240°30′. The FB of BC = 140°20′. Included angle at B = BB of AB − FB of BC = 240°30′ − 140°20′ = 100°10′.

Local attraction: Deviation of magnetic needle due to local magnetic influences (iron ore, electric cables). Detected when FB + BB ≠ 180° (for the same line). Corrections are applied.

▸Plane Table Survey (Introduction)

Purpose: To plot the map directly in the field. The plane table is a drawing board mounted on a tripod. The surveyor sights objects and draws rays on the paper.
Advantages: No field book needed, errors can be checked immediately, suitable for small areas. **Disadvantages:** Not suitable for large areas, cannot be used in rainy/hot weather, less accurate than theodolite surveys.

> **Module I summary:** Chain surveying is the simplest method for small areas. Compass survey uses bearings and angles; local attraction must be corrected. Plane table is a direct plotting method.

MODULE II

Levelling – Principles, Instruments and Reduction

16 hrs | CO2 | Bloom: Understand, Apply

Purpose of levelling, definitions (level surface, datum, reduced level, bench mark). Levelling instruments – dumpy, tilting, automatic. Staff types. Temporary adjustments. Level book format. Reduction of levels by Height of Collimation and Rise & Fall methods.

▸Level Surface, Datum, Reduced Level, Bench Mark

Level surface: A surface parallel to the mean spheroid of the earth (e.g., still water surface). **Datum:** A reference surface or line (e.g., Mean Sea Level). **Reduced Level (RL):** Height of a point above or below the datum. **Bench mark (BM):** A fixed point of known RL. Types: GTS BM (Great Trigonometrical Survey), Permanent BM, Arbitrary BM, Temporary BM.

> **Malayalam support:** Level surface എന്നത് ഭൂമിയുടെ ഉപരിതലത്തിന് സമാന്തരമായ ഒരു പ്രതലം. Datum എന്നത് അളക്കുന്നതിനുള്ള അവലംബം. Reduced Level (RL) എന്നത് ഡാറ്റത്തിൽ നിന്നുള്ള ഉയരം.

▸Levelling Instruments – Dumpy, Tilting, Automatic

Dumpy level: The telescope is rigidly fixed to the vertical spindle; no tilting screw. **Tilting level:** The telescope can be tilted by a screw; used for precise work. **Automatic level:** Uses a compensator to keep the line of sight horizontal even if the instrument is slightly tilted.

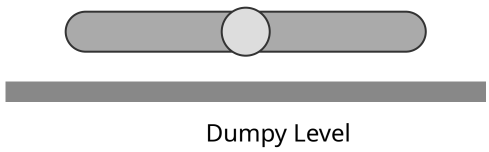


Fig: Dumpy level (schematic).

Levelling staff: A graduated rod held vertically to measure height differences. Types: self-reading staff (telescopic, folding) and target staff (for precise sighting).

▸Temporary Adjustments of a Levelling Instrument

1. **Setting up:** Place instrument on tripod, roughly level.
2. **Leveling:** Using foot screws, bring the bubble to the centre of its run.
3. **Focusing:** Focus the eyepiece for distinct vision of cross-hairs, then focus the objective to bring staff clearly.
4. **Eliminating parallax:** Ensure cross-hairs and staff image are in the same plane.

> 📌 Always check that the bubble is central before taking a reading.

▸Level Book – Format and Reduction of Levels

Level book: A table with columns: Station, Back Sight (BS), Intermediate Sight (IS), Fore Sight (FS), Rise, Fall, Reduced Level, Remarks.
Height of Collimation (HI) method: HI = RL of BM + BS. Then RL = HI − (IS or FS). **Rise & Fall method:** Rise = BS − IS (or FS), Fall = IS (or FS) − BS. RL of new point = RL of previous point + Rise (or − Fall).

> **Worked example (HI method):** BM RL=100.00 m, BS=1.50, IS=2.10, FS=0.80.
> HI = 100.00 + 1.50 = 101.50.
> RL at IS = 101.50 − 2.10 = 99.40 m. RL at FS = 101.50 − 0.80 = 100.70 m.

> **Worked example (Rise & Fall):** BS=1.50, IS=2.10 ⇒ Fall = 2.10 − 1.50 = 0.60 ⇒ RL = 100.00 − 0.60 = 99.40 m.
> Next FS=0.80 ⇒ Rise = 1.50 − 0.80 = 0.70 ⇒ RL = 100.00 + 0.70 = 100.70 m.

Feature	HI Method	Rise & Fall
Ease of computation	Faster	More tedious
Check	ΣBS − ΣFS = Last RL − First RL	ΣRise − ΣFall = Last RL − First RL
Error detection	Less direct	Better for identifying mistakes

Comparison of HI and Rise-Fall methods

> **Module II summary:** Levelling determines elevations. HI method is quicker; Rise-Fall provides better checks. Always keep the level book neat and apply arithmetic checks.

MODULE III

Levelling Applications and Contour Survey – Basics

16 hrs | CO3 | Bloom: Understand, Apply

Types of levelling (fly, profile, check, reciprocal). Numerical problems on RL computation, gradient calculations. Longitudinal and cross sections. Contour definition, characteristics, interval, and uses.

▸Fly Levelling, Profile Levelling, Check Levelling, Reciprocal Levelling

Fly levelling: Rapid levelling to establish bench marks, with fewer intermediate sights. **Profile levelling:** Determines ground elevations along a line (e.g., for road, pipeline). **Check levelling:** Re-levelling to verify previous work. **Reciprocal levelling:** Used when two points are far apart or obstructed; take readings from both ends to eliminate collimation error.

> **Malayalam support:** Fly levelling എന്നത് വേഗത്തിൽ bench marks നിർണ്ണയിക്കാൻ ഉപയോഗിക്കുന്നു. Profile levelling റോഡ്, പൈപ്പ്ലൈൻ തുടങ്ങിയവയുടെ സമീപത്തെ ഉയരം അളക്കുന്നു.

▸Numerical Problems on Levelling

Problems involve computing RLs using HI or Rise-Fall, applying checks, and calculating gradient between points.

> **Worked example:** The following readings were taken on a continuous sloping ground: 0.85, 1.25, 1.95, 2.45 (all IS), 1.20 (FS), 0.90 (BS). RL of BM = 100.00 m. Compute RLs using HI method.
> HI = 100.00 + 0.90 = 100.90.
> RLs: 100.90−0.85=100.05; 100.90−1.25=99.65; 100.90−1.95=98.95; 100.90−2.45=98.45; 100.90−1.20=99.70.
> Check: ΣBS−ΣFS = 0.90−1.20 = −0.30; Last RL − First RL = 99.70 − 100.00 = −0.30 ✓.

Gradient: The rate of rise or fall per unit horizontal distance. Gradient = (Difference in RL) / (Horizontal distance).

▸Longitudinal Section (L-Section) and Cross Section (C-Section)

L-section: A profile of the ground surface along a centre line (e.g., road alignment). **C-section:** Cross-sections perpendicular to the centre line, showing ground levels at intervals.
Plotting: L-section is plotted with distances on X-axis and RL on Y-axis. C-sections are drawn at each chainage, showing the ground shape.

▸Contour – Definition, Characteristics, Interval, Uses

Contour: An imaginary line joining points of equal elevation on the ground. **Characteristics:** (1) Contours do not intersect each other (except in vertical cliffs). (2) Closely spaced contours indicate steep slope; widely spaced indicate gentle slope. (3) Contours close to form circles around hills or depressions.

Contour interval: The vertical distance between successive contours. It depends on the scale and terrain.
Uses of contour maps: Determine topography, estimate earthwork, design roads/dams, locate drainage, etc.

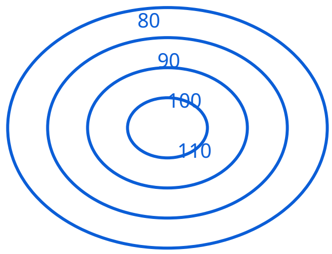


Fig: Contours of a hill (elevations in m).

Module III summary: Different levelling methods suit different purposes. Gradient calculations are essential for engineering design. Contour maps are powerful tools for terrain analysis.

MODULE IV

Introduction to Theodolite Surveying

12 hrs | CO4 | Bloom: Remember, Understand

Introduction to theodolite: types, main parts, temporary and permanent adjustments. Measurement of horizontal angles by general procedure, repetition, and reiteration. Applications in civil engineering.

▸Theodolite – Introduction and Types

A **theodolite** is a precision instrument for measuring horizontal and vertical angles.
Types: Transit theodolite (telescope can be revolved through 360° in vertical plane) and Non-transit theodolite. Also classified by size (e.g., 10″, 20″, 30″ – seconds of arc).

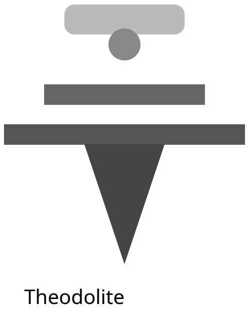


Fig: Schematic of a transit theodolite.

▸Main Parts of a Transit Theodolite

- **Telescope:** For sighting objects.
- **Vertical circle:** Measures vertical angles.
- **Horizontal circle:** Measures horizontal angles.
- **Plate bubbles:** For leveling.
- **Compass (optional):** For magnetic bearing.
- **Foot screws:** For leveling and centering.
- **Clamps and tangent screws:** For fine movements.

▸Temporary and Permanent Adjustments

Temporary adjustments (at each setup): Setting up, leveling, focusing, eliminating parallax, centering over the station.
Permanent adjustments: Fundamental relationships between the axes (e.g., vertical axis should be truly vertical, horizontal axis horizontal, line of sight perpendicular to horizontal axis). These are checked and adjusted at regular intervals.

▸Measurement of Horizontal Angles

General procedure: Set up theodolite over station, level it, sight target A (set horizontal circle to 0°), then sight target B, read the angle.
Method of Repetition: The angle is measured several times (e.g., 3–6 times) by repeating the sighting, and the average is taken. This reduces errors.
Method of Reiteration: Several angles at a station are measured by sighting all targets from a common zero, reading the circle each time. The angles are obtained by differences.

Worked example (Repetition): To measure angle AOB, set zero on A, sight B, read θ_1. Repeat: set zero on B, sight A, read θ_2. The angle = $(\theta_1 + \theta_2)/2$ (if θ_2 is $360° - \theta_1$). Practice problems.

▸Applications of Theodolite in Civil Engineering

- Setting out alignments for roads, railways, bridges.
- Measuring horizontal and vertical angles.
- Traverse surveys.
- Construction layout and alignment.
- Topographic mapping.

Module IV summary: The theodolite is a versatile instrument for precise angle measurement. Proper adjustments and careful procedure ensure accurate results.

📝 Formula Bank

Chain Survey:
Area (Trapezoidal) = $d \times [(O_0+O_n)/2 + \Sigma O_i]$

Compass:
BB = FB ± 180° (if FB<180°, +; if >180°, −)

Included angle:
∠ = BB(prev) − FB(next) (or adjusted)

HI method:
HI = RL + BS; RL = HI − IS/FS

Rise-Fall:
Rise = BS − FS/IS; Fall = FS/IS − BS

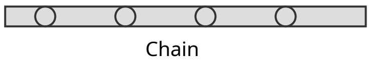
Check:
ΣBS − ΣFS = Last RL − First RL

Gradient:
G = ΔRL / Horizontal distance

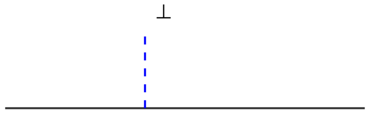
Contour interval:
CI = (RL difference) / (number of spaces)

🖼️ Visual Library

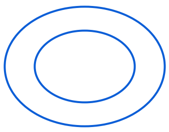
Chain



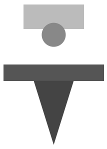
Offset



Contours



Theodolite



📋 Self-Learning Activities

Module I (Weeks 1-4): Draw labelled diagrams of chain, tape, compass; practice area calculations; prepare notes on bearings and included angles.

Module II (Weeks 5-8): Prepare glossary of levelling terms; study temporary adjustments; solve HI and Rise-Fall problems.

Module III (Weeks 9-12): Study fly, profile, reciprocal levelling; practice gradient calculations; sketch L-sections and contours.

Module IV (Weeks 13-15): Draw theodolite parts; write step-wise notes on horizontal angle measurement methods; study applications.

📊 Assessment & Examination Pattern

Mode	Attendance	CA1	CA2	CA3	CA4	CA5	CA6	ESE (Theory)
Marks	5	15	15	15	15	20	20	60

CIE and ESE Components

Series Examination Pattern (2 hours): Part A: 2 questions (1 mark each), Part B: 2 questions (3 marks each), Part C: 2 out of 3 questions (7 marks each) – Total 25 marks per series exam (2 such exams = 50 marks, converted to 40 CIE).
End Semester Examination (2.5 hours): Part A: 2 questions per module (1 mark each) = 8 marks; Part B: 2 out of 3 questions per module (3 marks each) = 24 marks; Part C: 1 set question with choice per module (7 marks each) = 28 marks; Total 60 marks.

❓ Module-wise Questions with Answers

Module I

▶**Define base line, tie line, and check line.**

Base line is the longest line through the centre of the survey; tie line joins two points on main lines; check line verifies accuracy.

▶**What is local attraction? How is it detected?**

Deviation of magnetic needle due to local influences. Detected when FB + BB ≠ 180°.

Module II

▶**Explain the Height of Collimation method with a numerical example.**

HI = RL + BS; RL = HI − IS/FS. Example: BM RL=100, BS=1.5, IS=2.1 ⇒ HI=101.5, RL=99.4 m.

Module III

▶**What is reciprocal levelling? Where is it used?**

Used when two points are far apart or obstructed. Readings are taken from both ends to eliminate collimation error.

Module IV

▶**Describe the method of repetition for measuring horizontal angle.**

The angle is measured repeatedly (3–6 times) by sighting the targets alternately, and the average is taken to reduce errors.

📄 Practice Model Paper

Based on the official pattern, not an official SITTTR model question paper.
Part A (1 mark each – any 8 of 8): (1) Define datum. (2) What is a tie line? (3) What is the formula for area by trapezoidal rule? (4) Define whole circle bearing. (5) What is a level surface? (6) What is a contour interval? (7) Name two types of levelling. (8) Name two parts of a theodolite.
Part B (3 marks each – any 2 of 3 per module): (Module I) Explain direct and indirect ranging. (Module II) Differentiate HI and Rise-Fall methods. (Module III) Explain the characteristics of contours. (Module IV) Describe temporary adjustments of a theodolite.
Part C (7 marks each – 1 set question with choice per module): (Module I) Calculate area from offsets. (Module II) Reduce levels using HI method. (Module III) Solve levelling numerical. (Module IV) Explain horizontal angle measurement by repetition.

⚡ Quick Revision

M1: Chain & tape; base/tie/check lines; offsets; bearings; included angle; local attraction; plane table intro.

M2: Level surface, datum, RL, BM; dumpy/automatic level; staff; HI and Rise-Fall; checks.

M3: Fly, profile, reciprocal levelling; gradient; L/C sections; contour characteristics & uses.

M4: Theodolite parts; temporary adjustments; horizontal angle – repetition & reiteration.

Common mistakes: Forgetting to check $\Sigma BS - \Sigma FS$ = Last RL − First RL; incorrect bearing corrections for local attraction; mixing up HI and Rise-Fall formulas; not adjusting theodolite properly.

Exam checklist: Practice numerical problems from each module; memorise definitions and key concepts; draw neat diagrams; manage time in exams.

Gradient Calculator

RL_1 (m): 100.0

RL_2 (m): 105.5

Distance (m): 50.0

Gradient = 0.1100 (rise per meter)

📖 Glossary

Term	Meaning
Base line	Longest line in a survey
Bearing	Angle between a line and a meridian
BM (Bench mark)	Point of known reduced level
Contour	Line joining equal elevations

Datum	Reference surface for heights
Gradient	Rate of change of elevation
HI (Height of Collimation)	Elevation of line of sight
RL (Reduced Level)	Height above datum
Theodolite	Instrument for measuring angles

📚 References

Textbooks: Surveying and Levelling – T.P. Kanetkar & S.V. Kulkarni; Surveying Vol. I & II – B.C. Punnia.
References: A Textbook of Surveying and Levelling – R. Agor; Surveying and Levelling – R. Subramanian.
Web Resources: NPTEL (Levelling, Contours, Theodolite); civilenggforall.com (Chain survey, Compass); engineeringcivil.com (All).